

Technology Stack

ComicCraft - AI Comic Story Creator using Gemini Models

Madras Institute of Technology | Department of Computer Technology (CT) | III Year, V Semester

Project period: 1 Oct 2026 - 2 Oct 2026

Layer	Technology	Purpose
Language	Python 3.14	Application code
Web framework	FastAPI + Uvicorn	Routes, forms, JSON API
Templates	Jinja2	Home, preview, export pages
Outline AI	Google Gemini Flash (google-generativeai)	Structured JSON outline
Story AI	Google Gemini Pro family	Captions and narration
Image AI	Stable Diffusion v1.5, diffusers, PyTorch, transformers, accelerate	Panel images (MPS/CUDA/CPU)
PDF	fpdf2 + DejaVu fonts	One page per panel
Imaging	Pillow	Saving images
Config	python-dotenv	GEMINI_API_KEY, model names
Frontend	HTML, CSS, Bangers and Poppins fonts	Responsive glass UI
Concurrency	concurrent.futures ThreadPoolExecutor	Overlap story and image generation

Rationale

- FastAPI: simple, typed, automatic docs at /docs
- Stable Diffusion: local, no per-image cost
- Gemini free tier lets anyone start without billing